

GUAM VTOL EOMS

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1. EQUATIONS OF MOTION

$$(1) \quad X^\top = \begin{bmatrix} \left(\mathbf{p}_{B/E}^E\right)^\top & \Phi^\top & \left(\mathbf{v}_{B/E}^B\right)^\top & \left(\omega_{B/E}^B\right)^\top \end{bmatrix}$$

$$(2) \quad \dot{\mathbf{p}}_{B/E}^E = C_{B/E} \mathbf{v}_{B/E}^B$$

$$(3) \quad \dot{\Phi} = H(\Phi) \omega_{B/E}^B$$

$$(4) \quad \dot{\mathbf{v}}_{B/E}^B = \frac{1}{m_B} \mathbf{F}^B + C_{B/E} \mathbf{g}^E - \omega_{B/E}^B \times \mathbf{v}_{B/E}^B$$

$$(5) \quad \dot{\omega}_{B/E}^B = J_B^{-1} \left(\mathbf{M}^B - \omega_{B/E}^B \times J \omega_{B/E}^B \right)$$

2. FORCES AND MOMENTS

We have

$$\mathbf{F}^B = \mathbf{F}_{motor}^B + \mathbf{F}_{aero}^B$$

$$\mathbf{M}^B = \mathbf{M}_{motor}^B + \mathbf{M}_{aero}^B$$

3. AERODYNAMICS

$$(6) \quad \mathbf{F}_{aero}^B = \frac{1}{2} \rho \cdot |\mathbf{v}|^2 \cdot S \cdot \begin{bmatrix} C_L(\mathbf{v}, \Phi, \mathbf{e}, \mathbf{a}, \mathbf{r}) \\ C_D(\mathbf{v}, \Phi, \mathbf{e}, \mathbf{a}, \mathbf{r}) \\ C_Q(\mathbf{v}, \Phi, \mathbf{e}, \mathbf{a}, \mathbf{r}) \end{bmatrix}$$

$$(7) \quad \mathbf{M}_{aero}^B = \frac{1}{2} \rho \cdot |\mathbf{v}|^2 \cdot S \cdot c \cdot \begin{bmatrix} C_m(\mathbf{v}, \Phi, \mathbf{e}, \mathbf{a}, \mathbf{r}) \\ C_{LA}(\mathbf{v}, \Phi, \mathbf{e}, \mathbf{a}, \mathbf{r}) \\ C_{NA}(\mathbf{v}, \Phi, \mathbf{e}, \mathbf{a}, \mathbf{r}) \end{bmatrix}$$

4. MOTORS

$$(8) \quad \mathbf{F}_{motor} = \begin{bmatrix} T_{push} \\ 0 \\ -\sum_i T_{lift,i} \end{bmatrix}$$

$$\text{w/ Thrust} := T_i = K_T \omega_i^2$$

$$(9) \quad \mathbf{M}_{motor} = \begin{bmatrix} \sum_i T_i dy_i + \tau_{push} \\ \sum_i T_i dx_i - T_{push} dz_{push} \\ \sum_i \tau_i \end{bmatrix}$$

$$\text{w/ Torque} := \tau_i = b\omega_i^2 + I_m \dot{\omega}_i$$

$$(10) \quad \mathbf{F}_{si}^W = L_{si} \mathbf{n}_{si}^B - \begin{bmatrix} D_{si} \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{F}_{si}^B = C_{B/W} \mathbf{F}_{si}^W$$

$$\mathbf{F}_{aero}^B = \sum_i \mathbf{F}_{si}^B$$

$$\mathbf{M}_{aero}^B = \sum_i \mathbf{M}_{si}^B$$

$$\mathbf{M}_{si}^B = M_{si} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \mathbf{n}_{si}^B + \mathbf{x}_{si/cm}^B \times \mathbf{F}_{si}^B$$

For each aero surface si with normal vector $\mathbf{n}_{si}^B$ and center of lift relative to center of mass $\mathbf{x}_{si/cm}$

$$L_{si} = \frac{1}{2} \rho \cdot |\mathbf{v}|^2 \cdot S_{si} \cdot C_L(\alpha_{si}, \delta u_i)$$

$$D_{si} = \frac{1}{2} \rho \cdot |\mathbf{v}|^2 \cdot S_{si} \cdot C_D(\alpha_{si}, \delta u_i)$$

$$M_{si} = \frac{1}{2} \rho \cdot |\mathbf{v}|^2 \cdot S_{si} \cdot c \cdot C_M(\alpha_{si}, \delta u_i)$$

$$\alpha_{si} = -\sin \left(\frac{\mathbf{v}_{B/W}^B \cdot \mathbf{n}_{si}^B}{|\mathbf{v}_{B/W}^B|} \right)$$

$$C_{L,si}(\alpha, \delta u) = C_{L0,si} + C_{L\alpha,si} \alpha + \left(\frac{\partial C_{L,si}}{\partial \delta u_i} \right) \delta u \left(\frac{S_{u,si}}{S_{si}} \right)$$

$$C_{D,si}(\alpha, \delta u) = C_{D0,si} + C_{D\alpha,si} (\alpha - \alpha_{0,si})^2 + \left(\frac{\partial C_{D,si}}{\partial \delta u_i} \right) \delta u^2 \left(\frac{S_{u,si}}{S_{si}} \right)$$

$$C_{M,si}(\alpha, \delta u) = C_{M0,si} + C_{M\alpha,si} \alpha + \left(\frac{\partial C_{M,si}}{\partial \delta u_i} \right) \delta u \left(\frac{S_{u,si}}{S_{si}} \right)$$